

# Does Vowel Diacritization Hurt LLM Question Answering? Evidence, Failure Modes, and Metric Artifacts in Hebrew and Arabic

Ameer Ganem amir.ganem@mail.huji.ac.il Elyasaf Nachum  
elyasaf.nachum@mail.huji.ac.il Yuval Fried yuval.fried@mail.huji.ac.il

## Abstract

Hebrew and Arabic are typically written without vowel diacritics (Niqud / Tashkeel), which appear only in formal, liturgical, or disambiguation-critical text. We ask whether adding diacritics to QA inputs helps or hurts LLMs, testing three models across six benchmarks in both languages, with datasets vocalized using SOTA automatic vocalizers. Our results show that diacritization consistently degrades Arabic QA accuracy, while in Hebrew, the raw signal looked noisy: Niqud produced a positive delta on one dataset but a negative delta on the others. Re-scoring with an LLM-as-judge sharpened the signal—and closer inspection revealed that the effect is not uniform even in Arabic, but rather a mix of model-specific causes. A causal ablation that changes only the Unicode placement of the diacritic mark, while holding its content fixed, substantially reduces both this collapse and the accuracy drop—evidence that tokenization placement, rather than diacritic presence alone, drives the damage. Throughout the paper, we work toward classifying the diacritic-induced failure modes we observe across models. We release a corpus of judge-confirmed diacritic-induced failure cases for future mechanistic analysis.

## 1 Introduction

Hebrew and Arabic are usually written without the small marks that spell out vowels (Niqud in Hebrew, Tashkeel in Arabic); readers infer the vowels from context. These marks appear precisely where correct interpretation matters most: children’s books, religious texts, poetry, and other disambiguation-critical writing. We ask whether adding these vowel marks to a passage, before it is given to a language model, changes the model’s ability to answer questions about that passage.

To our knowledge, no prior published work directly studies the effect of vowel diacritization on LLMs for Hebrew and Arabic QA. At first glance, we found only a weak correlation between model failure on vocalized QA and subword fragmentation ( $r = 0.13$ ), based on dataset-level F1/Exact

Match (EM) scores. These results suggested that the effect of diacritization cannot be explained by subword fragmentation alone, motivating a more fine-grained analysis of its impact on model behavior.

We revisit the question at the per-example level with three additions: **a causal ablation that varies only the Unicode placement of the diacritic marks** while holding its content fixed, isolating tokenization as a mechanism rather than inferring it from correlation; **an LLM-as-judge re-scoring pass** that tests whether F1/EM’s conclusions hold under a semantic, rather than lexical, correctness check; and **manual inspection** of individual failure and success cases across both languages. It turns out that vocalization does not produce one uniform effect but a range of model-specific failure modes — and occasional genuine improvements — which we try to classify, e.g. a fixed stock-phrase collapse resembling sentiment classification (SILMA, Arabic), a shift toward code-switching into Chinese (Qwen, Hebrew), and — surprisingly — shorter, more accurate answers (dictalm2, Hebrew). The LLM-as-judge does not replace this classification effort; it sharpens it by giving us a more reliable correctness signal to classify against.

Any system serving Hebrew or Arabic readers of diacritized text — religious, educational, or literary content among them — risks a real accuracy degradation that standard benchmark scores can obscure or even contradict.

Our contributions: (i) we classify the model-specific failure — and occasional success — modes that diacritization induces across Hebrew and Arabic, including collapse into fixed, task-irrelevant stock answers, cross-lingual code-switching, and unexpected answer-shortening effects; (ii) we show that this failure is not simply driven by the presence of diacritics: an ablation that holds diacritic content fixed and changes only its Unicode placement disproportionately shrinks both the collapse rate and the accuracy drop, ruling out diacritic presence alone as the explanation,

though the precise mechanism behind the placement effect remains an open question; (iii) we mine and release 898 judge-confirmed diacritic-induced failure cases as a resource for follow-up interpretability work (logit-lens / patchscopes probing of entity retrieval). As part of the study, we also diacritized all six evaluated datasets and release the resulting versions alongside our code, enabling reproduction and further investigation of the observed effects.

## 2 Data

We evaluate on three QA benchmarks per language, all drawn from existing, publicly available, human-authored benchmarks rather than data we collected or generated ourselves. **Arabic:** ARCD (Mozannar et al., 2019) ( $n=1,395$ ), TyDiQA-Arabic (Clark et al., 2020) ( $n=15,726$ ), and MKQA-Arabic (Longpre et al., 2021). **Hebrew:** HeQ (Cohen et al., 2023) ( $n=8,363$ ), ParaShoot (Keren and Levy, 2021) ( $n=3,038$ ), and MKQA-Hebrew (Longpre et al., 2021) ( $n=4,573$ ). Together, these benchmarks span extractive QA and reading comprehension, with open-domain QA.

For each dataset we construct a paired evaluation: the identical example is presented to the model with and without diacritics, so that any accuracy difference is attributable to the diacritics alone. Diacritics were added automatically, not by human annotators: Hebrew Niqud was applied with DICTA’s dictabert-large-charmenaked (Shmidman et al., 2020), and Arabic Tashkeel with Farasa (Abdelali et al., 2016). Table 1 reports per-dataset sizes.

| Dataset   | Language | n      |
|-----------|----------|--------|
| ARCD      | Arabic   | 1,395  |
| TyDiQA    | Arabic   | 15,726 |
| MKQA-AR   | Arabic   | 4,966  |
| HeQ       | Hebrew   | 8,363  |
| ParaShoot | Hebrew   | 3,038  |
| MKQA-HE   | Hebrew   | 4,573  |

Table 1: Per-dataset example counts.

## 3 Methods

**Models.** Arabic: SILMA-9B-Instruct (SILMA.AI, 2024) (a model natively fine-tuned for Arabic), Qwen2.5-7B-Instruct (Team, 2024), and dictalm2.0-instruct (Shmidman et al., 2024) (a model fine-tuned for Hebrew, used here

as a cross-lingual control on Arabic). Hebrew: Qwen2.5-7B-Instruct (Team, 2024) and dictalm2.0-instruct (Shmidman et al., 2024) (fine-tuned natively for Hebrew).

**Primary metric.** Token-level F1 and exact match (EM) between the model’s generated answer and the gold span, computed separately for the plain and diacritized condition and reported as  $\Delta = \text{diacritized} - \text{plain}$ .

**Fragmentation ratio.** For each model/dataset we compute the ratio of subword-tokenizer token count on diacritized vs. plain text, as a proxy for how much diacritics inflate the input’s effective length.

**Postfix ablation.** Standard Unicode representation places a combining diacritic mark *before* the base-letter’s tokenizer boundary in a way that inflates fragmentation; we generate an alternative encoding that reorders combining marks to a canonical postfix position relative to tokenizer merges, changing only how the identical diacritic content is byte-encoded, and re-run SILMA on the same examples.

**Collapse mining.** We frequency-rank each model’s diacritized-condition predictions to detect degenerate, task-irrelevant stock answers repeated far more often than plain-condition predictions.

**LLM-as-judge.** Because F1/EM penalize any lexical deviation from the gold span even when an answer is semantically correct (or credit partial n-gram overlap even when an answer is semantically wrong), we additionally re-score a stratified sample (500–600 examples per dataset/model, matched pairs across the plain/diacritized conditions) with gpt-4o-mini-2024-07-18 (OpenAI), accessed via the OpenAI API, acting as a binary CORRECT/INCORRECT judge given the question, gold answer(s), and model prediction; the same judge model and prompt were used for both the Hebrew and Arabic samples. We report judge accuracy and  $\Delta$  alongside F1/EM.

## 4 Results

### 4.1 Arabic results

Under standard (prefix) Tashkeel, every model degrades on every dataset (Table 2); dictalm2 — a Hebrew-tuned model with no native Arabic pre-training, used here as a cross-lingual control — takes the largest F1 hit despite having the smallest tokenizer fragmentation of the three (Table 4).

| Model      | Dataset        | EM <sub>plain</sub> | EM <sub>Tash</sub> | ΔEM          | F1 <sub>plain</sub> | F1 <sub>Tash</sub> | ΔF1          |
|------------|----------------|---------------------|--------------------|--------------|---------------------|--------------------|--------------|
| Qwen2.5-7B | ARCD           | 27.46               | 21.36              | -6.09        | 64.66               | 58.90              | -5.76        |
| Qwen2.5-7B | TyDiQA         | 22.73               | 18.85              | -3.88        | 56.96               | 50.80              | -6.16        |
| Qwen2.5-7B | MKQA           | 2.15                | 1.55               | -0.60        | 6.97                | 6.11               | -0.87        |
| Qwen2.5-7B | <b>Overall</b> | <b>18.40</b>        | <b>15.12</b>       | <b>-3.28</b> | <b>46.21</b>        | <b>41.26</b>       | <b>-4.95</b> |
| SILMA-9B   | ARCD           | 41.86               | 37.28              | -4.59        | 70.52               | 64.01              | -6.51        |
| SILMA-9B   | TyDiQA         | 48.78               | 40.55              | -8.23        | 68.06               | 61.69              | -6.37        |
| SILMA-9B   | MKQA           | 6.56                | 4.33               | -2.24        | 12.12               | 7.83               | -4.29        |
| SILMA-9B   | <b>Overall</b> | <b>38.85</b>        | <b>32.20</b>       | <b>-6.65</b> | <b>55.64</b>        | <b>49.73</b>       | <b>-5.91</b> |
| dictalm2   | ARCD           | 11.83               | 8.89               | -2.94        | 48.46               | 39.96              | -8.50        |
| dictalm2   | TyDiQA         | 16.49               | 13.41              | -3.08        | 48.34               | 40.19              | -8.15        |
| dictalm2   | MKQA           | 1.15                | 0.50               | -0.64        | 2.90                | 2.28               | -0.62        |
| dictalm2   | <b>Overall</b> | <b>12.75</b>        | <b>10.22</b>       | <b>-2.53</b> | <b>38.13</b>        | <b>31.65</b>       | <b>-6.48</b> |

Table 2: Arabic: EM/F1, plain vs. Tashkeel, per model and dataset ( $n=22,087$  total). Every  $\Delta$  is negative.

## 4.2 Hebrew results

Under Niqud, the raw picture is mixed rather than uniform: both models gain on HeQ but lose on ParaShoot and MKQA (Table 3) — the source of the “no effect” reading the pooled aggregate invites, examined in Section 4.5.

| Model | Ds         | EM <sub>P</sub> | EM <sub>N</sub> | ΔEM          | F1 <sub>P</sub> | F1 <sub>N</sub> | ΔF1          |
|-------|------------|-----------------|-----------------|--------------|-----------------|-----------------|--------------|
| Qwen  | HeQ        | 10.09           | 11.99           | +1.90        | 16.68           | 20.35           | +3.68        |
| Qwen  | PS         | 27.49           | 19.32           | -8.16        | 58.04           | 44.88           | -13.15       |
| Qwen  | MKQA       | 1.70            | 0.94            | -0.77        | 5.37            | 3.34            | -2.03        |
| Qwen  | <b>All</b> | <b>11.00</b>    | <b>10.22</b>    | <b>-0.78</b> | <b>21.31</b>    | <b>20.15</b>    | <b>-1.16</b> |
| Dicta | HeQ        | 7.29            | 11.97           | +4.68        | 22.64           | 27.10           | +4.46        |
| Dicta | PS         | 10.63           | 10.37           | -0.26        | 47.63           | 41.54           | -6.09        |
| Dicta | MKQA       | 8.31            | 1.90            | -6.41        | 16.90           | 7.39            | -9.51        |
| Dicta | <b>All</b> | <b>8.22</b>     | <b>8.78</b>     | <b>+0.56</b> | <b>25.75</b>    | <b>24.20</b>    | <b>-1.55</b> |

Table 3: Hebrew: EM/F1, plain vs. Niqud, per model and dataset ( $n=15,974$  total). PS = ParaShoot.

## 4.3 Why the raw signal diverges

Arabic Tashkeel degrades every model on every dataset; Hebrew Niqud is noisier, helping on HeQ while hurting ParaShoot and MKQA. Fragmentation ratio does not explain either pattern: Table 4 shows the *lowest*-fragmentation Arabic model (dictalm2) taking the *largest* hit, and §4.5 shows Hebrew and Arabic converge to a similar magnitude once judge-scored, despite Hebrew’s far higher fragmentation ratios for the same architectures — fragmentation ratio is not the driver in either language.

| Model (Arabic) | Frag. ratio | ΔF1   |
|----------------|-------------|-------|
| dictalm2       | 1.53–1.55×  | -6.48 |
| SILMA-9B       | 2.54–2.62×  | -5.91 |
| Qwen2.5-7B     | 2.67–2.81×  | -4.95 |

Table 4: Fragmentation ratio (plain→Tashkeel token inflation) vs. overall ΔF1, ranked ascending. Pearson  $r=0.13$  across all model×dataset pairs.

## 4.4 Failure Modes, Recovery Cases, and Deeper Inspection of Diacritization Effects

### SILMA: sentiment-phrasal collapse (Arabic).

Under Tashkeel, SILMA collapses onto a fixed stock phrase on 5.5% of ARCD+TyDiQA questions (0.5% under plain text,  $\sim 11\times$ ), answering unrelated factual questions as if performing sentiment classification rather than QA:

**Q:** ما هي الهندسة الميكانيكية ؟ (“What is mechanical engineering?”) — **Gold:** a branch of engineering concerned with design, manufacture, and operation of machines — **Pred:** هو رأي سلبي (“it is a negative opinion”)  
**Q:** من هم الرونن ؟ (“Who are the ronin?”) — **Gold:** a samurai without a lord — **Pred:** هو رأي سلبي (“it is a negative opinion”)

Postfix Unicode re-encoding (Table 5) — identical diacritic content, different tokenizer-boundary placement — nearly halves both the collapse rate and the damage, implicating *placement*, not diacritics per se.

| SILMA (Arabic) | Prefix     | Postfix                      |
|----------------|------------|------------------------------|
| ΔF1            | -6.39      | -3.84                        |
| ΔEM            | -8.04      | -5.57                        |
| Collapse rate  | 5.43%      | 2.29%                        |
| Frag. ratio    | 2.54–2.62× | $\sim 13\text{--}14\%$ lower |

Table 5: Diacritic Unicode placement alone (content unchanged) vs. SILMA’s Arabic degradation and collapse rate (plain-baseline collapse rate  $\approx 0.5\%$  in both).

### Qwen: code-switch collapse (Hebrew, HeQ).

On HeQ, 23.3% (140/600) of Qwen’s *plain*-condition predictions degrade into a Chinese-language refusal — more than double the 10.8% (65/600) rate under Niqud — and not one of the 140 is ever judged correct:

**Q:** GETT? באיזו שנה הוקמה (“In what year was GETT founded?”) — **Gold:** 2013

Plain: 质 safegטע GETT, הוקמת, 1981  
 228 给你们提供的文本并没有提到GETT的成立年份...  
 229 ("...the passage does not state GETT's founding  
 230 year...") Niqud: 2013 (correct)

Q: מה היא החברה הרווחית ביותר בכל הזמנים?  
 231 ("What is the most profitable company of all  
 232 time?") — Gold: אפל (Apple)  
 233 Plain: התשובה אינה מופיעה  
 234 בקטע所提供的段落并没有提到全球最赚钱的公司相关信息...  
 235 ("...the passage does not mention the world's  
 236 most profitable company...") Niqud: אפל  
 237 ("Apple," correct)  
 238

239 This runs *opposite* to SILMA's collapse — it  
 240 fires *without* diacritics — and is HeQ-specific  
 241 (ParaShoot: 14/500 vs. 12/500; MKQA: 19/400 vs.  
 242 29/400, both small and symmetric). It is also the  
 243 full explanation for Qwen's apparent Niqud "help"  
 244 on HeQ (§4.5).

245 **dictalm2: shorter, more accurate answers**  
 246 **(Hebrew, HeQ).** Under Niqud, dictalm2's HeQ an-  
 247 swers get shorter and land on the gold span more  
 248 often; EM rewards only exact spans, so a terser reg-  
 249 ister lifts EM whenever the plain-condition answer  
 250 wraps the identical fact in a full sentence:

251 Q: מתי החלה תנועת המחאה האפרו-אמריקאית?  
 252 ("When did the Afro-American protest  
 253 movement begin?") — Gold: 1965  
 254 Niqud: 1965 (EM=1) Plain: תנועת המחאה  
 255 האפרו-אמריקאית החלה ב-1965.  
 256 ("...began in 1965." EM=0 — right content, wrong form)

257 Q: כמה עלינו לשתות ביום? ("How much should  
 258 we drink per day?") — Gold: 4 בקבוקי שתייה  
 259 (4 bottles)  
 260 Niqud: 4 בקבוקי שתייה (EM=1) Plain: אנחנו  
 261 צריכים לשתות לפחות 3 מחסניות של Cirkul ביום.  
 262 ("...at least 3 Cirkul cartridges..." EM=0,  
 263 wrong content)

264 We initially attributed the apparent HeQ gain to  
 265 the shorter, more compact answers dictalm2 pro-  
 266 duces under Niqud (Section 4.4, "shorter, more ac-  
 267 curate answers"): EM rewards a terse span that  
 268 lands exactly on the gold answer, while the plain-  
 269 condition answer often wraps the same fact in a full  
 270 sentence and is penalized despite being equally cor-  
 271 rect. Re-scoring with the LLM-as-judge, however,  
 272 reveals that this explanation only partly survives:  
 273 for dictalm2, the judge-based gap on HeQ nearly  
 274 flattens (+4.46 F1 → +1.17 judge-Δ), confirm-  
 275 ing that most of the raw gain was a surface-form  
 276 artifact of EM/F1 rewarding brevity rather than  
 277 genuine comprehension. For Qwen, the picture is  
 278 different again: its HeQ "gain" turns out to be al-  
 279 most entirely a byproduct of the code-switch-into-  
 280 Chinese collapse (Section 4.4) — a failure mode

that, for reasons that remain unclear, fires *less* of-  
 ten once the input is vocalized.

**Association-driven substitution.** A word or  
 proper noun inside the *vocalized* question itself  
 pulls in an unrelated but strongly co-occurring en-  
 tity, distinct from confusable substitution in that  
 the wrong answer is traceable to a specific lexical  
 trigger in the prompt rather than mere domain prox-  
 imity:

Q: מי היה הזמר המקורי של השיר הללויה? ("Who  
 was the original singer of the song  
 Hallelujah?") — Gold: Leonard Cohen —  
 Pred: דוד המלך ("King David" — pulled in via  
 the biblical/Psalms association of the vocalized  
 word הללויה)

Q: מי נלחם בקרב קדש? ("Who fought at the  
 Battle of Kadesh?") — Gold: Ramesses II —  
 Pred: משה ("Moses" — pulled in via the Torah  
 association of the vocalized place name קדש,  
 rather than the historical battle it names)

**General pattern: subject echo.** Diacritics  
 sometimes trigger the model to answer with the en-  
 tity *named in the question* instead of retrieving the  
 fact associated with it:

Q: איזו חברה משתמשת בשירותיה של  
 BrandTotal? ("Which company uses  
 BrandTotal's services?") — Gold: פייסבוק  
 (Facebook)  
 Plain: פייסבוק (correct) Niqud: BrandTotal  
 (echoes the queried entity itself)

Q: מי הזמר של system of a down? ("Who is the  
 singer of System of a Down?") — Gold: סרג'  
 טנקיאן (Serj Tankian)  
 Plain: סרג' טנקיאן (correct) Niqud: of System  
 of a Down ("System of a Down is the  
 singer.")

More general patterns can be found in Ap-  
 pendix B.

Both are mined from the 898 judge-confirmed  
 harm pairs (§5) and motivate the logit-lens follow-  
 up: does the correct entity remain linearly decod-  
 able before the model collapses onto the echo?

## 4.5 LLM-judge validation

**Arabic.** The judge (stratified sample, same  
 three datasets) finds a uniformly larger effect than  
 F1/EM but preserves the exact cross-model rank-  
 ing — dictalm2 > SILMA > Qwen (Table 6) —  
 validating F1/EM's relative ordering even though  
 it understates the absolute size.

| Model      | F1/EM $\Delta$ | Judge $\Delta$ |
|------------|----------------|----------------|
| dictalm2   | -6.48 / -2.53  | -8.13          |
| SILMA-9B   | -5.91 / -6.65  | -7.46          |
| Qwen2.5-7B | -4.95 / -3.28  | -6.27          |

Table 6: Arabic overall  $\Delta$  (F1/EM vs. judge), ranked by judge magnitude.

**Hebrew.** The judge confirms F1/EM understated ParaShoot/MKQA harm (Table 7); its apparent HeQ "help" is Qwen's code-switch-collapse artifact (§4.4), not evidence Niqud aids comprehension — only dictalm2's HeQ gain is genuine. Corrected for this artifact, Hebrew's judge-based effect (-5.67 Qwen, -8.93 dictalm2, combined  $n=3,000$ ,  $\Delta=-7.3$ ) is the same order of magnitude as Arabic's.

| Model    | Ds   | F1 $\Delta$ | Judge $\Delta$ |
|----------|------|-------------|----------------|
| Qwen     | HeQ  | +3.68       | +3.5           |
| Qwen     | PS   | -13.15      | -17.8          |
| Qwen     | MKQA | -2.03       | -4.25          |
| Qwen     | All  | -1.16       | -5.67          |
| dictalm2 | HeQ  | +4.46       | +1.17          |
| dictalm2 | PS   | -6.09       | -16.6          |
| dictalm2 | MKQA | -9.51       | -14.5          |
| dictalm2 | All  | -1.55       | -8.93          |

Table 7: Hebrew per-dataset  $\Delta$ : F1 vs. judge. PS = ParaShoot.

## 5 Discussion, Limitations, and Future Work

The LLM-judge sample (500–600 examples/dataset/model) is smaller than the full F1/EM evaluation and was scored by a single judge model without a human inter-rater agreement check; we treat its deltas as a validation signal for metric reliability rather than a replacement ground truth. A Qwen postfix-Tashkeel rerun — which would test whether the tokenization-placement effect generalizes beyond SILMA — remains future work. We have mined 898 judge-confirmed correct→incorrect pairs (427 Hebrew, 471 Arabic) and manually curated a subset with short, extractable entity answers (company/person/place names) as candidates for a logit-lens / patchscopes probe of whether the correct entity remains linearly decodable from intermediate hidden states even when the diacritized-condition output is wrong, motivated by the subject-echo pattern in Section 4.4. This mechanistic analysis is in progress and not yet part of this report's results.

## 6 Conclusion

Vowel diacritization measurably degrades LLM QA accuracy in both Hebrew and Arabic, by a similar magnitude once evaluated with a metric robust to lexical-surface noise. In Arabic this degradation is metric-robust and partly explained by a model-specific stock-phrase collapse (SILMA) that a tokenization-only intervention (postfix diacritic placement) nearly halves, implicating tokenization as a causal — not merely correlated — driver. In Hebrew, raw F1/EM aggregates are unreliable for two compounding reasons: they nearly cancel to zero due to opposing per-dataset effects and partial-credit scoring, and Qwen's one apparently helpful dataset (HeQ) is itself a collapse artifact — a code-switch-into-Chinese failure mode that fires more often *without* diacritics, not evidence that diacritics help. Once both artifacts are accounted for, Hebrew shows a real, consistent degradation on the same order as Arabic's, with dictalm2's HeQ improvement standing out as a genuine, still-unexplained exception. Practically, this argues for (a) caution when using surface-overlap metrics to evaluate any morphologically rich, diacritic-sensitive language, (b) diacritic-placement-aware tokenization as a concrete, low-cost mitigation worth further study, and (c) checking apparent "diacritics help" signals for collapse artifacts before trusting them.

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## A LLM-as-Judge Prompts

Both judges use gpt-4o-mini-2024-07-18, a temperature of 0, and a strict JSON output schema. The system prompt is written in English in both languages; only the worked examples and the language-specific spelling/diacritic exception differ. The Arabic and Hebrew judge runs are otherwise identical pipelines (blinded to model identity and experimental condition, no F1/EM signal passed in).

### A.1 Hebrew judge

You are a Hebrew QA evaluation judge. Your task is to assess whether a candidate answer correctly answers a question, given the reference (gold) answer(s).

Judge the candidate based on whether it sufficiently and factually answers the question relative to the provided gold answer(s).

If a context passage is provided, use it as additional evidence. If no context is provided, judge based on the question and gold answer(s). Treat the provided gold answer(s) as the reference; do not override them using outside knowledge.

You must respond with valid JSON only. No explanation outside the JSON. The value of "label" must be exactly one of "CORRECT" or

"INCORRECT". Use exactly this JSON structure: {"label": "CORRECT", "reason": "one short sentence explaining your judgment"}

**CORRECT:**

- The candidate correctly and sufficiently answers what the question asks.
- It does NOT need to reproduce the entire gold answer if the shorter answer fully answers the question.
- Harmless additional information is allowed.
- Different wording, word order, spelling variants, transliteration, punctuation, capitalization, or formatting are allowed when the meaning is unchanged.
- An equivalent answer in another language or script is allowed when it unambiguously refers to the same entity, value, date, place, or concept.
- Do not penalize Hebrew niqud (vowel diacritics) being present or absent.
- Do not penalize ktiv haser vs. ktiv male spelling differences (defective vs. full spelling).
- Do not penalize harmless grammatical or morphological differences, including prefixes such as the definite article.
- Equivalent date, number, unit, or title formatting is allowed.
- If multiple gold answers are provided, the candidate is CORRECT if it correctly matches any valid gold answer.

**Examples:**

Gold: "בשנת 2002" Candidate: "2002" → CORRECT

Gold: "ה-24" Candidate: "יום 24" → CORRECT

Gold: "ירושלים" Candidate: "יְרוּשָׁלַיִם" → CORRECT

**INCORRECT:**

- The core answer is factually wrong.
- The candidate does not answer the question.
- The candidate contains a material factual error or contradiction, even if part of the answer is correct.
- Merely being semantically related to the gold answer is not sufficient.

**Important rules:**

- Judge correctness, not lexical overlap.

-- Judge whether the QUESTION has been sufficiently answered, not whether every word in the gold answer was reproduced.

-- A shorter answer is CORRECT when it contains all information necessary to answer the question.

-- Extra information is allowed only when it does not introduce a material error.

-- Do not use F1, EM, model identity, or experimental condition in your judgment.

User message template: ``Question: {question}\n\nGold answer(s): {gold\_answers}\n\nCandidate answer: {prediction}``

## A.2 Arabic judge

You are an Arabic QA evaluation judge. Your task is to assess whether a candidate answer correctly answers a question, given the reference (gold) answer(s).

Judge the candidate based on whether it sufficiently and factually answers the question relative to the provided gold answer(s).

If a context passage is provided, use it as additional evidence. If no context is provided, judge based on the question and gold answer(s). Treat the provided gold answer(s) as the reference; do not override them using outside knowledge.

You must respond with valid JSON only. No explanation outside the JSON. The value of "label" must be exactly one of "CORRECT" or "INCORRECT". Use exactly this JSON structure: {"label": "CORRECT", "reason": "one short sentence explaining your judgment"}

CORRECT:

-- The candidate correctly and sufficiently answers what the question asks.

-- It does NOT need to reproduce the entire gold answer if the shorter answer fully answers the question.

-- Harmless additional information is allowed.

-- Different wording, word order, spelling variants, transliteration, punctuation, capitalization, or formatting are allowed when the meaning is unchanged.

-- An equivalent answer in another language or script is allowed when it unambiguously refers to the same

entity, value, date, place, or concept.

-- Do not penalize Arabic diacritics (Tashkeel) being present or absent.

-- Do not penalize harmless grammatical/case-ending differences.

-- Equivalent date, number, unit, or title formatting is allowed.

-- If multiple gold answers are provided, the candidate is CORRECT if it correctly matches any valid gold answer.

Examples:

Gold: "2002 في عام" Candidate: "2002" → CORRECT

Gold: "1999م" Candidate: "1999" → CORRECT

Gold: "مَسْقُط" Candidate: "مَسْقُط" → CORRECT

INCORRECT:

-- The core answer is factually wrong.

-- The candidate does not answer the question.

-- The candidate contains a material factual error or contradiction, even if part of the answer is correct.

-- Merely being semantically related to the gold answer is not sufficient.

Important rules:

-- Judge correctness, not lexical overlap.

-- Judge whether the QUESTION has been sufficiently answered, not whether every word in the gold answer was reproduced.

-- A shorter answer is CORRECT when it contains all information necessary to answer the question.

-- Extra information is allowed only when it does not introduce a material error.

-- Do not use F1, EM, model identity, or experimental condition in your judgment.

User message template: ``Question: {question}\n\nGold answer(s): {gold\_answers}\n\nCandidate answer: {prediction}``

## B Additional Diacritization Failure-Mode Patterns

Beyond the subject-echo pattern reported in Section 4.4, manual inspection of the 898 judge-confirmed correct→incorrect pairs surfaces several further recurring patterns, mined primarily from entity-answer questions in HeQ and MKQA

(Hebrew) where the gold answer is a short<sup>12</sup> check-able name. We report them here as candidates for the same logit-lens / patchscopes follow-up described in Section 5, since — like subject echo — each raises a distinct hypothesis about *where* in the network the correct entity is lost.

**Entity-to-generic-noun blank.** The correct proper noun is dropped in favor of a generic category noun, as though the model retained the semantic *type* of the answer (a person, a title) without the specific referent:

**Q:** למי יש הכי הרבה מדליות זזהב בעולם ("Who has the most Olympic gold medals in the world?") — **Gold:** Michael Phelps — **Pred:** למה יש הרבה מדליות זזהב בעולם ("the king has many gold medals in the world")

**Confusable-entity substitution (same category).** The model names a real entity of the correct type and domain, but the wrong instance — swapping in another member of the same franchise, award category, or competition:

**Q:** "Who played Snape in the Harry Potter films?" — **Gold:** Alan Rickman — **Pred:** Rupert Grint (a different Harry Potter actor; playing Ron, not Snape)

**Q:** "Who won the Academy Award for Best Actress in 2018?" — **Gold:** Frances McDormand — **Pred:** Jennifer Lawrence (a different Best-Actress winner; wrong year)

**Full/unrelated hallucination.** The answer is a real, well-formed entity but has no discernible topical link to the question — a stronger failure than confusable substitution, since even the answer’s category or domain is wrong:

**Q:** איזו מדינה אירופית שלטה באחד דרום אפריקה ("Which European country ruled the Union of South Africa?") — **Gold:** Britain — **Pred:** אלג'יריה ("Algeria" — not even a colonial power on that continent)

**Category-break hallucination.** The predicted answer switches *answer type* entirely, not just referent — e.g. substituting an invented personal name where the question unambiguously calls for a country:

**Q:** מי זכה במונדיאל בשנת 1958 ("Who won the World Cup in 1958?") — **Gold:** Brazil — **Pred:** פרנץ בורנשטיין (an invented person’s name, though the question asks which country won a team competition)

**Numeric/format degradation.** The correct magnitude is retrieved but rendered in a corrupted

surface form — sign-flipped, unit-dropped, or otherwise malformed — suggesting the failure is localized to answer *formatting* rather than fact retrieval:

**Q:** כמה פעמים בודקים בודקיים בודקיים בודקיים? ("How many times does DoppelPaymer check each request?") — **Gold:** 3 times — **Pred:** “-3” (correct magnitude, sign-flipped, unit word dropped)

Taken together with subject echo (Section 4.4), these six patterns span a spectrum from pure surface-format corruption (numeric/format degradation) through partial retrieval (entity-to-generic-noun blank, subject echo) to full semantic substitution (confusable and association-driven substitution) to total retrieval failure (full and category-break hallucination) — motivating layer-wise probing across this spectrum rather than treating diacritic-induced harm as a single failure type.